For this next exercise, you will be analyzing TSA complaint data. There are 4 datasets attached. 3 provide information on the complaints, counts by Airport, counts by category, and the final dataset includes the airport name by code. For this exercise, you must craft a story about the data based on the following parameters: audience, purpose, medium, design, and ethical considerations.

To get started, analyze the data and determine what looks interesting.

Requirements:

1. You must use the 4 datasets provided for whatever medium you are choosing to present your story and remember it must include a call to action.
2. This exercise can use Tableau or Python.
3. You must define an audience, purpose, and medium to tell your story.
4. You must have at least 6 visuals included that use the data provided and 3 must be from this list:
- Heat Maps
- Custom Visuals
- Box Plots
- Spatial Visuals / Choropleth Maps
5. Include a 250-500-word paper summarizing your audience, purpose, medium and design choices along with the above ethical consideration questions.

In [1]:
```
import pandas as pd
import numpy as np

airport = pd.read_csv('complaints-by-airport.csv')
category = pd.read_csv('complaints-by-category.csv')
subcategory = pd.read_csv('complaints-by-subcategory.csv')
iata = pd.read_csv('iata-icao.csv')
```

In [2]:
```
print(airport.info())
airport.head()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 41721 entries, 0 to 41720
Data columns (total 4 columns):
 #   Column           Non-Null Count  Dtype 
---  ------           --------------  ----- 
 0   pdf_report_date  41721 non-null  object
 1   airport          41612 non-null  object
 2   year_month       41721 non-null  object
 3   count            41721 non-null  int64 
dtypes: int64(1), object(3)
memory usage: 1.3+ MB
None
```

Out[2]:

	pdf_report_date	airport	year_month	count
0	2019-02	ABE	2015-01	0
1	2019-02	ABE	2015-02	0
2	2019-02	ABE	2015-03	0
3	2019-02	ABE	2015-04	0
4	2019-02	ABE	2015-05	2

In [3]:
```
print(category.info())
category.head()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 241588 entries, 0 to 241587
Data columns (total 7 columns):
 #   Column            Non-Null Count   Dtype 
---  ------            --------------   ----- 
 0   pdf_report_date   241588 non-null  object
 1   airport           237546 non-null  object
 2   category          241588 non-null  object
 3   year_month        241588 non-null  object
 4   count             241588 non-null  int64 
 5   clean_cat         241588 non-null  object
 6   clean_cat_status  241588 non-null  object
dtypes: int64(1), object(6)
memory usage: 12.9+ MB
None
```

Out[3]:

	pdf_report_date	airport	category	year_month	count	clean_cat	clean_cat_st
0	2019-02	ABE	Hazardous Materials Safety	2015-01	0	Hazardous Materials Safety	ori
1	2019-02	ABE	Mishandling of Passenger Property	2015-01	0	Mishandling of Passenger Property	ori
2	2019-02	ABE	Hazardous Materials Safety	2015-02	0	Hazardous Materials Safety	ori
3	2019-02	ABE	Mishandling of Passenger Property	2015-02	0	Mishandling of Passenger Property	ori
4	2019-02	ABE	Hazardous Materials Safety	2015-03	0	Hazardous Materials Safety	ori

In [4]:
```
print(subcategory.info())
subcategory.head()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 504512 entries, 0 to 504511
Data columns (total 11 columns):
 #   Column                     Non-Null Count   Dtype 
---  ------                     --------------   ----- 
 0   pdf_report_date            504512 non-null  object
 1   airport                    491048 non-null  object
 2   category                   504512 non-null  object
 3   subcategory                504512 non-null  object
 4   year_month                 504512 non-null  object
 5   count                      504512 non-null  int64 
 6   clean_cat                  504512 non-null  object
 7   clean_subcat               504512 non-null  object
 8   clean_cat_status           504512 non-null  object
 9   clean_subcat_status        504512 non-null  object
 10  is_category_prefix_removed  504512 non-null  bool  
dtypes: bool(1), int64(1), object(9)
memory usage: 39.0+ MB
None
```

Out[4]:

	pdf_report_date	airport	category	subcategory	year_month	count	clean_cat
0	2019-02	ABE	Hazardous Materials Safety	General	2015-01	0	Hazardous Materials Safety
1	2019-02	ABE	Mishandling of Passenger Property	Damaged/ Missing Items--Checked Baggage	2015-01	0	Mishandling of Passenger Property
2	2019-02	ABE	Hazardous Materials Safety	General	2015-02	0	Hazardous Materials Safety
3	2019-02	ABE	Mishandling of Passenger Property	Damaged/ Missing Items--Checked Baggage	2015-02	0	Mishandling of Passenger Property
4	2019-02	ABE	Hazardous Materials Safety	General	2015-03	0	Hazardous Materials Safety

In [5]:
```
print(iata.info())
iata.head()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8937 entries, 0 to 8936
Data columns (total 7 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   country_code  8905 non-null   object 
 1   region_name   8937 non-null   object 
 2   iata          8937 non-null   object 
 3   icao          7794 non-null   object 
 4   airport       8937 non-null   object 
 5   latitude      8937 non-null   float64
 6   longitude     8937 non-null   float64
dtypes: float64(2), object(5)
memory usage: 488.9+ KB
None
```

Out[5]:

	country_code	region_name	iata	icao	airport	latitude	longitude
0	AE	Abu Zaby	AAN	OMAL	Al Ain International Airport	24.2617	55.6092
1	AE	Abu Zaby	AUH	OMAA	Abu Dhabi International Airport	24.4330	54.6511
2	AE	Abu Zaby	AYM	NaN	Yas Island Seaplane Base	24.4670	54.6103
3	AE	Abu Zaby	AZI	OMAD	Al Bateen Executive Airport	24.4283	54.4581
4	AE	Abu Zaby	DHF	OMAM	Al Dhafra Air Base	24.2482	54.5477

In [6]:
```
iata['country_code'].unique()
```

Out[6]:
```
array(['AE', 'AF', 'AG', 'AI', 'AM', 'AO', 'AQ', 'AR', 'AS', 'AT', 'AU',
       'AW', 'AZ', 'BA', 'BB', 'BD', 'BE', 'BF', 'BG', 'BH', 'BI', 'BJ',
       'BL', 'BM', 'BN', 'BO', 'BQ', 'BR', 'BS', 'BT', 'BW', 'BY', 'BZ',
       'CA', 'CC', 'CD', 'CF', 'CG', 'CH', 'CI', 'CK', 'CL', 'CM', 'CN',
       'CO', 'CR', 'CU', 'CV', 'CW', 'CX', 'CY', 'CZ', 'DE', 'DJ', 'DK',
       'DM', 'DO', 'DZ', 'EC', 'EE', 'EG', 'ER', 'ES', 'ET', 'FI', 'FJ',
       'FK', 'FM', 'FR', 'GA', 'GB', 'GD', 'GE', 'GF', 'GH', 'GI', 'GL',
       'GM', 'GN', 'GP', 'GQ', 'GR', 'GT', 'GU', 'GW', 'GY', 'HK', 'HN',
       'HR', 'HT', 'HU', 'ID', 'IE', 'IL', 'IN', 'IQ', 'IR', 'IS', 'IT',
       'JE', 'JM', 'JO', 'JP', 'KE', 'KG', 'KH', 'KI', 'KM', 'KN', 'KP',
       'KR', 'KW', 'KY', 'KZ', 'LA', 'LB', 'LC', 'LK', 'LR', 'LS', 'LT',
       'LU', 'LV', 'LY', 'MA', 'MD', 'ME', 'MG', 'MH', 'MK', 'ML', 'MM',
       'MN', 'MO', 'MP', 'MQ', 'MR', 'MS', 'MT', 'MU', 'MV', 'MW', 'MX',
       'MY', 'MZ', nan, 'NC', 'NE', 'NF', 'NG', 'NI', 'NL', 'NO', 'NP',
       'NR', 'NU', 'NZ', 'OM', 'PA', 'PE', 'PF', 'PG', 'PH', 'PK', 'PL',
       'PM', 'PR', 'PS', 'PT', 'PW', 'PY', 'QA', 'RE', 'RO', 'RS', 'RU',
       'RW', 'SA', 'SB', 'SC', 'SD', 'SE', 'SG', 'SH', 'SJ', 'SK', 'SL',
       'SN', 'SO', 'SR', 'SS', 'ST', 'SV', 'SX', 'SY', 'SZ', 'TC', 'TD',
       'TG', 'TH', 'TJ', 'TL', 'TM', 'TN', 'TO', 'TR', 'TT', 'TV', 'TW',
       'TZ', 'UA', 'UG', 'UM', 'US', 'UY', 'UZ', 'VC', 'VE', 'VG', 'VI',
       'VN', 'VU', 'WF', 'WS', 'YE', 'YT', 'ZA', 'ZM', 'ZW'], dtype=object)
```

In [ ]:

In [ ]:

In [7]:
```
#filter IATA to US only and rename to match airport column
us_iata = (
    iata[iata["country_code"] == "US"]
    [["iata", "airport", "latitude", "longitude", "region_name"]]
    .rename(columns = {"iata": "airport", "airport": "airport_name", "region

#total complaints per airport, then join coordinates
merged = (
    airport
    .groupby("airport", as_index = False)["count"].sum()
    .rename(columns = {"count": "total_complaints"})
    .merge(us_iata, on = "airport", how = "inner"))

#merged.to_csv("merged_choropleth_airports.csv", index = False)
merged.head()
```

Out[7]:

	airport	total_complaints	airport_name	latitude	longitude	state
0	ABE	463	Lehigh Valley International Airport	40.6521	-75.4408	Pennsylvania
1	ABI	109	Abilene Regional Airport	32.4113	-99.6819	Texas
2	ABQ	2478	Albuquerque International Sunport	35.0402	-106.6090	New Mexico
3	ABR	60	Aberdeen Regional Airport	45.4491	-98.4218	South Dakota
4	ABY	118	Southwest Georgia Regional Airport	31.5355	-84.1945	Georgia

In [ ]:

In [8]:
```
iata = (
    iata[iata["country_code"] == "US"]
    [["iata", "airport", "region_name"]]
    .rename(columns = {"iata": "airport", "airport": "airport_name", "region

#filter to the two equity categories and pivot to wide format
equity_cats = ["Civil Rights", "Persons w/ Disabilities (PWD)"]

merged2 = (
    category[category["clean_cat"].isin(equity_cats)]
    .groupby(["airport", "clean_cat"], as_index = False)["count"].sum()
    .pivot_table(index = "airport", columns = "clean_cat", values = "count",
    .reset_index())

merged2.columns.name = None
merged2 = merged2.rename(columns={"Civil Rights": "civil_rights", "Persons w
```

```
#add total complaints for bubble sizing
totals = (airport.groupby("airport")["count"].sum().reset_index().rename(col

merged2 = (merged2.merge(totals, on = "airport", how = "left").merge(iata, o

#merged2.to_csv("merged2_equity_airports.csv", index = False)
merged2.head()
```

Out[8]:

	airport	civil_rights	pwd_complaints	total_complaints	airport_name	state
0	ABE	12.0	9.0	463	Lehigh Valley International Airport	Pennsylvania
1	ABI	5.0	4.0	109	Abilene Regional Airport	Texas
2	ABQ	64.0	69.0	2478	Albuquerque International Sunport	New Mexico
3	ABR	3.0	1.0	60	Aberdeen Regional Airport	South Dakota
4	ABY	2.0	5.0	118	Southwest Georgia Regional Airport	Georgia

In [ ]: